

Effect of a 21 day Daniel Fast on metabolic and cardiovascular disease risk factors in men and women

Background Dietary modification via caloric restriction is associated with multiple effects related to improved metabolic and cardiovascular health. However, a mandated reduction in kilocalories is not well-tolerated by many individuals, limiting the long-term application of such a plan. The Daniel Fast is a widely utilized fast based on the Biblical book of Daniel. It involves a 21 day ad libitum food intake period, devoid of animal products and preservatives, and inclusive of fruits, vegetables, whole grains, legumes, nuts, and seeds. The purpose of the present study was to determine the efficacy of the Daniel Fast to improve markers of metabolic and cardiovascular disease risk. Methods 43 subjects (13 men; 30 women; 35 ± 1 yrs; range: 20-62 yrs) completed a 21 day period of modified food intake in accordance with detailed guidelines provided by investigators. All subjects purchased and prepared their own food. Following initial screening, subjects were given one week to prepare for the fast, after which time they reported to the lab for their pre-intervention assessment (day 1). After the 21 day fast, subjects reported to the lab for their post-intervention assessment (day 22). For both visits, subjects reported in a 12 hr fasted state, performing no strenuous physical activity during the preceding 24-48 hrs. At each visit, mental and physical health (SF-12 form), resting heart rate and blood pressure, and anthropometric variables were measured. Blood was collected for determination of complete blood count, metabolic panel, lipid panel, insulin, HOMA-IR, and C-reactive protein (CRP). Subjects' self-reported compliance, mood, and satiety in relation to the fast were also recorded. Diet records were maintained by all subjects during the 7 day period immediately prior to the fast (usual intake) and during the final 7 days of the fast. Results Subjects' compliance to the fast was $98.7 \pm 0.2\%$ (mean $\pm$ SEM). Using a 10 point scale, subjects' mood and satiety were both 7.9 ± 0.2 . The following variables were significantly ($p < 0.05$) lower following the fast as compared to before the fast: white blood cell count (5.68 ± 0.24 vs. $4.99 \pm 0.19 \times 10^3/\mu\text{L}$), blood urea nitrogen (13.07 ± 0.58 vs. $10.14 \pm 0.59 \text{ mg}\cdot\text{dL}^{-1}$), blood urea nitrogen/creatinine (14.74 ± 0.59

vs. 11.67 ± 0.68), protein (6.95 ± 0.07 vs. 6.77 ± 0.06 g·dL⁻¹), total cholesterol (171.07 ± 4.57 vs. 138.69 ± 4.39 mg·dL⁻¹), LDL-C (98.38 ± 3.89 vs. 76.07 ± 3.53 mg·dL⁻¹), HDL-C (55.65 ± 2.50 vs. 47.58 ± 2.19 mg·dL⁻¹), SBP (114.65 ± 2.34 vs. 105.93 ± 2.12 mmHg), and DBP (72.23 ± 1.59 vs. 67.00 ± 1.43 mmHg). Insulin (4.42 ± 0.52 vs. 3.37 ± 0.35 μ U·mL⁻¹; $p = 0.10$), HOMA-IR (0.97 ± 0.13 vs. 0.72 ± 0.08 ; $p = 0.10$), and CRP (3.15 ± 0.91 vs. 1.60 ± 0.42 mg·L⁻¹; $p = 0.13$), were lowered to a clinically meaningful, albeit statistically insignificant extent. No significant difference was noted for any anthropometric variable ($p > 0.05$). As expected, multiple differences in dietary intake were noted ($p < 0.05$), including a reduction in total kilocalorie intake (2185 ± 94 vs. 1722 ± 85). Conclusion A 21 day period of modified dietary intake in accordance with the Daniel Fast is 1) well-tolerated by men and women and 2) improves several risk factors for metabolic and cardiovascular disease. Larger scale, randomized studies, inclusive of a longer time period and possibly a slight modification in food choice in an attempt to maintain HDL cholesterol, are needed to extend these findings.